

Applying Unified Architecture Framework (UAF) for Systems of Systems Architectures

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Categorisation

- Accessibility: BEGINNER
- Application: GOOD PRACTICE
- Topics: Enterprise Systems Engineering, MBSE

Abstract

Architecture frameworks continue to evolve. Taking industry demand into account and addressing changing landscape of existing domain specific architecture frameworks, in September of 2013, a Request for Proposal for Unified profile for DoDAF and MODAF (UPDM 3.0) (later renamed to Unified Architecture Framework (UAF)) was created. Since the issue of the request for proposal (RFP), UPDM 3.0 working group identified the list of requirements. To name a few: Architecture Modeling Support for Defense, Industry, and government Organizations, support of Security Domain, Human Systems Integration, Support of Systems of Systems (SoS) Life Cycle Processes and Analyses. Since then four years past and a brand new UAF 1.0 became an official Object Management Group (OMG) standard and is about to become ISO/IEC 19517 standard. What is this new architecture framework (AF) all about? How is it different from other AFs? Is it simply a synthesis of a legacy? Or is it a modern technology combining the best practices of the old with a new and modern?

This paper explores UAF, answering the fundamental questions above and putting a specific focus on answering the questions below:

- What kind of problems is UAF applicable for solving?
- Is UAF a replacement for NATO architecture framework (NAF)?
- How is UAF related to model-based systems engineering (MBSE)?

These and many more questions are answered in this paper by presenting real-world UAF application examples of architecting systems of systems in various industry domains.

Introduction

Applying the Unified Architecture Framework, using an MBSE approach moves the architecture modelling effort to one that is an integral part of SE [Hause et al. 2016]. This helps the systems integrator develop interoperable systems with traceability to requirements and across views, using one integrated architecture model that enables impact analysis, gap analysis, trade studies, simulations, and engineering analysis [Morkevicius et al. 2017a]. Moreover, the scope of UAF is expanded beyond defense architectures. It is genericized to be applicable to architecting systems of systems of any domain. One of the mandatory requirements for UAF was “Architecture Modeling Support for Defense, Industry, and Government Organizations”. As a response to this requirement,

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UAF version 1.0 is industry domain agnostic [OMG 2013]. It is targeted to model systems of systems, including enterprises and IoT systems.

Why UAF? The paradigm shift from a document-centric systems engineering approach to model-based systems engineering (MBSE) revealed gaps in the MBSE approach, one of which was that no standardized methodology was available for SoS. The belief that Systems Modeling Language [OMG 2015] was the ultimate solution turned out to be incorrect. Modelling language by definition provides syntax and semantics, but not pragmatics. To apply a language such as SysML successfully, various questions must be answered, including how to structure the model, what views to build, which artefacts to deliver, and in what sequence. Every company deals with these issues differently. Organizations not complying with a standardized approach end up having differently structured models with different sets of views, resulting in loss of the capability to exchange data between models, loss of the capability to communicate with other teams, overhead in tool customization, and the need for specific training. Moreover, the models become impossible to integrate and reuse [Morkevicius et al. 2017a].

UAF

UAF consists of three main components (all UAF components are shown graphically in Figure 1):

- framework – a collection of domains, model kinds, and viewpoints,
- metamodel – a collection of types, tuples, and individuals used to construct views according to the specific viewpoints,
- profile – SysML based implementation of the metamodel to apply model-based systems engineering principles and best practices while building the views.

Two supplementary components are: (i) a traceability guide to other existing EAFs and UPDM; and (ii) an example model based on a search & rescue case study.

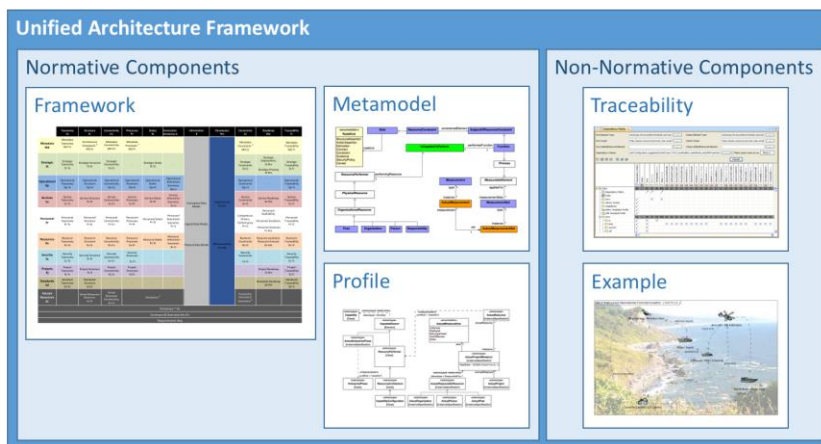


Figure 1 – UAF Framework Components

Framework

The Grid format (Figure 2) is the best to describe the framework component of the UAF. It is organized into rows and columns, where rows are Domains and columns are Model Kinds. The intersection of a row and column is called a Viewpoint. A UAF grid summarizes all viewpoints available in existing AFs. It serves as a foundation for building domain-specific frameworks by selecting only the viewpoints required for a specific context.

	Taxonomy Tx	Structure Sr	Connectivity Cn	Processes Pr	States St	Interaction Scenarios Is	Information If	Parameters Pm	Constraints Ct	Roadmap Rm	Traceability Tr
Metadata Md	Metadata Taxonomy Md-Tx	Architecture Viewpoints ^a Md-Sr	Metadata Connectivity Md-Cn	Metadata Processes ^a Md-Pr	-	-	Conceptual Data Model,	Environment Pm-En	Metadata Constraints ^a Md-Ct	-	Metadata Traceability Md-Tr
Strategic St	Strategic Taxonomy St-Tx	Strategic Structure St-Sr	Strategic Connectivity St-Cn	-	Strategic States St-St	-			Strategic Constraints St-Ct	Strategic Deployment, St-Rm Strategic Phasing St-Rm	Strategic Traceability St-Tr
Operational Op	Operational Taxonomy Op-Tx	Operational Structure Op-Sr	Operational Connectivity Op-Cn	Operational Processes Op-Pr	Operational States Op-St	Operational Interaction Scenarios Op-Is			Operational Constraints Op-Ct	-	Operational Traceability Op-Tr
Services Sv	Service Taxonomy Sv-Tx	Service Structure Sv-Sr	Service Connectivity Sv-Cn	Service Processes Sv-Pr	Service States Sv-St	Service Interaction Scenarios Sv-Is			Service Constraints Sv-Ct	Service Roadmap Sv-Rm	Service Traceability Sv-Tr
Personnel Pr	Personnel Taxonomy Pr-Tx	Personnel Structure Pr-Sr	Personnel Connectivity Pr-Cn	Personnel Processes Pr-Pr	Personnel States Pr-St	Personnel Interaction Scenarios Pr-Is	Logical Data Model,	Measurements Pm-Me	Competence, Drivers, Performance Pr-Ct	Personnel Availability, Personnel Evolution, Personnel Forecast Pr-Rm	Personnel Traceability Pr-Tr
Resources Rs	Resource Taxonomy Rs-Tx	Resource Structure Rs-Sr	Resource Connectivity Rs-Cn	Resource Processes Rs-Pr	Resource States Rs-St	Resource Interaction Scenarios Rs-Is	Physical Data Model		Resource Constraints Rs-Ct	Resource evolution, Resource forecast Rs-Rm	Resource Traceability Rs-Tr
Security Sc	Security Taxonomy Sc-Tx	Security Structure Sc-Sr	Security Connectivity Sc-Cn	Security Processes Sc-Pr	-	-	Security Constraints Sc-Ct		-	Security Traceability Sc-Tr	
Projects Pj	Project Taxonomy Pj-Tx	Project Structure Pj-Sr	Project Connectivity Pj-Cn	Project Processes Pj-Pr	-	-	-		Project Roadmap Pj-Rm	Project Traceability Pj-Tr	
Standards Sd	Standard Taxonomy Sd-Tx	Standards Structure Sd-Sr	-	-	-	-	-	-	Standards Roadmap Sd-Rm	Standards Traceability Sd-Tr	
Actuals Resources Ar		Actual Resources Structure, Ar-Sr	Actual Resources Connectivity, Ar-Cn	Simulation ^b				Parametric Execution/ Evaluation ^b	-	-	
Dictionary ^a Dc											
Summary & Overview Sm-Ov											
Requirements Req											

Figure 2 – UAF Grid

UAF consists of 13 Domains [OMG 2017]:

- Metadata - captures meta-data relevant to the entire architecture, e.g. principles, metamodel extensions, views to be built, processes of developing architecture, etc.
- Strategic - describes capability taxonomy, composition, dependencies and evolution.
- Operational - describes the requirements, operational behaviour, structure, and exchanges required to support (exhibit) capabilities.
- Services - shows Service Specifications and required and provided service levels of these specifications required to exhibit a Capability or to support an Operational Activity.
- Personnel - enables an understanding of the human role in systems/enterprise architectures. It provides a basis for decisions by stakeholders by providing a structured linkage from the engineering community to the manpower, personnel, training, and human factors communities.
- Resources - captures a solution architecture consisting of resources, e.g. organizational, software, artefacts, capability configurations, natural resources that implement the operational requirements.
- Security - illustrates the security assets, security constraints, security controls, families, and measures required to address specific security concerns.

- The intersection between rows and columns is called the viewpoint or view specification. It is a method to build a view. A View is an instance of a viewpoint [ISO 2011]. Every view can be expressed in different presentation artefacts, such as diagrams, tables, matrixes, graphs, charts, etc. UAF specification provides recommendations on presentation artefacts for every viewpoint. Recommended presentation artefacts are described in more detail in the Modelling Language section of the paper.

A UAF Domain metamodel (DMM) depicted in Figure 3 is organized according to viewpoints. Thus, it is easy to understand which elements (including types, individuals, and tuples) can be used to build a specific view. The categorization of elements into types, individuals, and tuples is taken from IDEAS. In general, a UAF metamodel is a simplified version of complex 4D IDEAS ontology [IDEAS 2012]. Although it is simplified, it is still powerful compared to the majority of existing enterprise modeling languages and methodologies.



A UAF metamodel uses color coding to distinguish between tuples, types, abstract types, and individuals. It is very helpful when reading the metamodel. The image above, however, depicts a different color coding. Element colors reflect row color in the grid. It simplifies the look of the metamodel, showing the main concepts and relationships defined in the UAF metamodel.

Profile

Profile, in the context of UML, defines limited extensions to a reference metamodel with the purpose of adapting the metamodel to a specific platform or domain. A UAF profile defines UML extensions to support a UAF metamodel. It is also dependent on a SysML profile, which is another extension to UML. Dependencies to SysML are in the form of inheritance relationships. The purpose of this inheritance is to inherit SysML graphical notation, and engineering analysis techniques applicable to SysML (e.g. parametric analysis).

A UAF Profile (UAFP) is organized in accordance with viewpoints, exactly the same as a UAF metamodel. With this organization, it is easy to understand what elements can be used to build a specific view. In addition to the metamodel, every profile element extends a specific UML metaclass and inherits from a specific SysML stereotype (e.g. Operational Performer extends a UML Class and inherits from a SysML block). A UAF profile also defines meta-constraints. For example, a Maps To capability relationship can only connect Activity to Capability.

UAF metamodel implementation as a UML profile provides several key benefits including:

- fUML standard can be used to execute UAF architectures
- SysML parametrics can be used to perform analytical engineering studies on UAF models
- UAF provides seamless integration with SysML, UML, and other OMG standard profiles to ensure traceability from high level systems of systems model to systems and software engineering models
- UAF is easily extendable

The key points listed above emphasize that the UAF profile is the recommended standard-based implementation of a UAF metamodel.

Modeling Language

A UAF metamodel does not define graphical notation. Systems Modeling Language (SysML) is the standard method of representing UAF metamodel elements graphically. SysML defines 9 diagram types, as well as notation for all SysML elements [OMG 2015]. Because the UAF profile is an extension of SysML, it inherits the same notation. For example, UAF Capability is inherited from SysML block, meaning that UAF Capability inherits SysML block notation.

UAF specification provides guidance for which SysML diagram type should be used to create a specific UAF view. It is important to note that it is only a guide. Other modeling languages (such as UML, BPMN, etc.) can also be used. In addition to SysML diagrams, it is recommended that some UAF views be represented in a tabular, matrix, or chart format.

UAFP, as an extension to SysML and UML, supports standard OMG interchange formats (XMI standard for model data interchange and DDI for diagram interchange). Additionally, UAF supports the concepts

of views and viewpoints, where a viewpoint specifies a method for building a specific view from model data dynamically.

Application Use Cases

There are already numerous applications of UAF. The defense industry is most commonly thought of; however, there are UAF applications in other industries that are well worth looking at.

The National environmental satellite, data, and information service (NESDIS) ground enterprise and NASA's Joint Polar Satellite Systems (JPSS) ground project are both successful systems of systems architecture examples by Jeffries Technology Solutions, Inc. Both were architected in DoDAF using a UPDM metamodel (the predecessor of UAFP). Lessons learned are that the MBSE approach reduces rework, increases accuracy, and enables robust and informed decision-making [Barnes 2018].

The development of large, complex, heavy construction equipment can be difficult, time consuming and expensive, even greater if the goal is to design a complete site solution. This example is taken from a real, ongoing project called Electric Site, carried out by Volvo Construction Equipment. The aim is to electrify a transport stage in a quarry, from excavation to primary crushing and transport to secondary crushing. This will reduce the CO2 by 95% and the total cost of operation by 25%. [Kihlström 2018] and [Sjöberg 2017] describe how a UAF model can be used to significantly influence continued system engineering efforts, as well as the software architecture for the application development. The enterprise architecture model has been developed specifically to ensure overall management of the site.

Other publicly available case studies are the Submarine Warfare Federated Tactical Systems (SWFTS) project by Lockheed Martin [Mitchell 2010], Airbus Helicopters [Wirtz 2017], UAF for IoT architectures [Morkevicius et al. 2017b].

UAF vs. NAF vs. DoDAF vs. TOGAF vs. Archimate

A UAF framework is an alternative to NAF and DoDAF. This statement, however, concerns the UAF framework component only. Other components supplement both DoDAF and NAF. For example, a UAF metamodel and profile are compatible with both NAF and DoDAF. Depending on the tools used to create DoDAF or NAF views, either the UAF metamodel or the profile can be used. Using the same metamodel for creating UAF, NAF, and DoDAF views enables interoperability between the frameworks. It is up to the tool vendor to provide a conversion mechanism to convert DoDAF OV-2 to NAF L2 or to UAF Operational Structure. The UAF naming scheme is much more intuitive than the ones provided by other frameworks. Because of this there are examples where UAF is used to develop architecture and the architecture is converted to DoDAF just before delivery.

UAF compatibility with TOGAF is different and is worth discussing separately. A UAF metamodel is an alternative to a TOGAF content metamodel. This means TOGAF architectures can be simply created using a UAF metamodel following the TOGAF architecture development method (ADM). The TOGAF ADM can also be combined with a UAF metamodel to build UAF views. This combination compensates for a lack of UAF architecture development method.

The UAF metamodel and Archimate are both named by the NATO 3C board as the official metamodels to create NAF views [Ristani 2018]. On one hand, they are competing alternatives. On the other hand, NATO requires OMG and the Open Group to find a way to support interoperability between the two. Initial ideas were developed in the OMG UAF group on how Archimate could work with UAF, i.e. using Archimate to develop high-level capability architectures and UAF to create rich operational and resource architectures.

Conclusion

UAF is the collection of best practices used in systems, systems of systems, and enterprise engineering for more than 30 years. Moreover, UAF is applicable to any domain. Additionally, is still the best choice to build DoDAF, NAF, and MODAF architectures.

A UAFP profile is a formal way of building UAF models. This formal background allows users to build precise, executable architectures, perform automated trade off analysis, run what-if scenarios, verify requirements, and add traceability to the systems and software models.

The applications of UAF, UAFP, and its predecessor UPDM has been proven in practical applications in different domains. There are multiple publicly-available case studies that prove UAF is the number one choice to build mature and rich systems of systems architectures.

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